THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	Each stage multiplies the count by 3: $3^1 = 3$
2	**(a)** 243	Each stage multiplies the count by 3: $3^5 = 243$
3	**(a)** 6	Count the faces
4	**(b)** 5	Count the faces
5	**(b)** 7	Count the faces
6	**(b)** 8	Euler's relation
7	**(a)** 4	Four segments of one-third length
8	**(d)** 1	Plus two circular faces
9	**(b)** circle	Looking from above
10	**(b)** 8	Corners

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A pattern that repeats itself at smaller scales (1); Sierpinski triangle or Koch snowflake (1).	1+1
12	$3^3 = 27$ (2).	2
13	A flat pattern that folds into the solid (1); six squares joined (1).	1+1
14	6, 12 and 8 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Number $= 3^1 = 3$ (2); side $= \frac{1}{2^1}$ unit (1).	2+1
16	Number $= 3^2 = 9$ (2); side $= \frac{1}{2^2}$ unit (1).	2+1
17	Number $= 3^3 = 27$ (2); side $= \frac{1}{2^3}$ unit (1).	2+1
	OR: Number $= 3^4 = 81$ (2); side $= \frac{1}{2^4}$ unit (1).	2+1
18	Number $= 3^5 = 243$ (2); side $= \frac{1}{2^5}$ unit (1).	2+1
	OR: Number $= 3^6 = 729$ (2); side $= \frac{1}{2^6}$ unit (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Remove the central inverted triangle from each triangle repeatedly (2). (ii) 1, 3, 9, 27 (1.5). (iii) Multiplied by 3 each stage (1.5).	2+1½+1½
	OR: (i) A solid bounded by plane polygonal faces (2). (ii) $6 + 8 - 12 = 2$; $5 + 5 - 8 = 2$ (3).	2+3
20	(i) 10×6 (2), 10×4 (2), 6×4 (2) (2). (ii) $2(60 + 40 + 24) = 248$ cm² (2). (iii) For gluing (1).	2+2+1
	OR: (i) Replace the middle third of each side by two segments forming a triangle (3). (ii) Side 1 cm, sides $3 \times 4 \times 4 = 48$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	Each stage multiplies the count by 3: $3^2 = 9$
2	**(c)** 729	Each stage multiplies the count by 3: $3^6 = 729$
3	**(c)** 12	Count the edges
4	**(c)** 8	Count the edges
5	**(a)** 15	Count the edges
6	**(b)** 18	Euler's relation
7	**(c)** net	Net of a solid
8	**(a)** lateral faces	Rectangles for a right prism
9	**(c)** cylinder	Has a curved surface
10	**(b)** regular tetrahedron	Four faces

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$F = 5, V = 6, E = 9$; $5 + 6 - 9 = 2$ (2).	2
12	A circle with a dot (centre) (2).	2
13	A cylinder (2).	2
14	5: one square and four triangles (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$F = 8, E = 18, V = 12$ (2); $F + V - E = 2 = 2$ (1).	2+1
16	$F = 9, E = 16, V = 9$ (2); $F + V - E = 2 = 2$ (1).	2+1
17	$F = 7, E = 12, V = 7$ (2); $F + V - E = 2 = 2$ (1).	2+1
	OR: $F = 7, E = 15, V = 10$ (2); $F + V - E = 2 = 2$ (1).	2+1
18	$F = 4, E = 6, V = 4$ (2); $F + V - E = 2 = 2$ (1).	2+1
	OR: $F = 10, E = 24, V = 16$ (2); $F + V - E = 2 = 2$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Top: rectangle; front: rectangle; side: circle (3). (ii) Identification (2).	3+2
	OR: (i) Cone (1). (ii) 2 faces (one curved, one flat), 1 edge (2). (iii) A sector and a circle (2).	1+2+2
20	(i) 60 (2). (ii) Inside cubes $3 \times 2 \times 1 = 6$; surface $= 54$ (3).	2+3
	OR: (i) A rectangle (3). (ii) A circle (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.